Hw11

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Question 1

a.

```
set.seed(43211)
x <- matrix(rnorm(10 * 2), nrow = 10, ncol = 2)
colnames(x) <- c("X1", "X2")

initial <- sample(rep(1:2, each = 5))
initial
```

```
##  [1] 2 1 1 2 2 2 1 1 2 1
```

```
means <- rbind(
  colMeans(x[initial == 1, , drop = FALSE]),
  colMeans(x[initial == 2, , drop = FALSE])
)
rownames(means) <- c("Cluster 1", "Cluster 2")
means
```

```
##                  X1         X2
## Cluster 1 0.3515306 -0.7937171
## Cluster 2 0.3079708  0.5240714
```

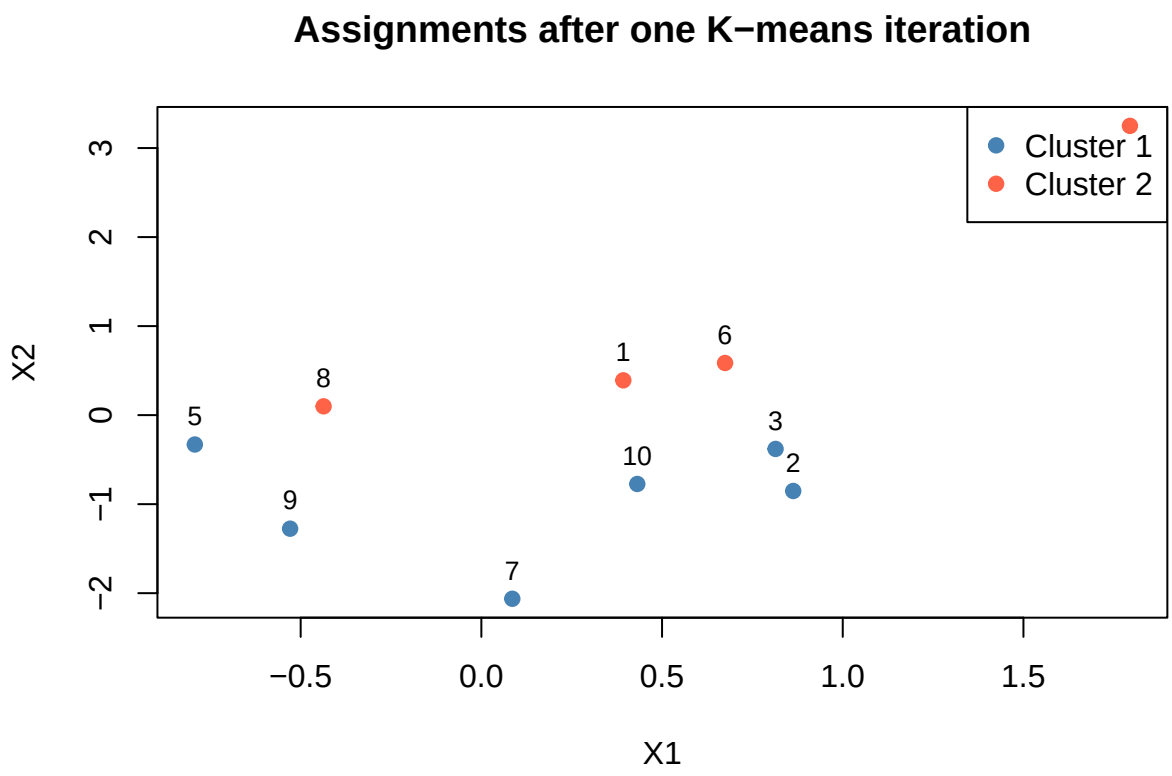
```
distance2 <- cbind(
  rowSums(sweep(x, 2, means[1, ], "-")^2),
  rowSums(sweep(x, 2, means[2, ], "-")^2)
)

updated <- apply(distance2, 1, which.min)
data.frame(observation = 1:10, initial, updated)
```

```
##    observation initial updated
## 1            1       2       2
## 2            2       1       1
## 3            3       1       1
## 4            4       2       2
## 5            5       2       1
## 6            6       2       2
## 7            7       1       1
```

```
## 8            8       1       2
## 9            9       2       1
## 10          10       1       1
```

```
colors <- c("steelblue", "tomato")
plot(x, col = colors[updated], pch = 19,
     xlab = "X1", ylab = "X2",
     main = "Assignments after one K-means iteration")
text(x, labels = 1:10, pos = 3, cex = 0.8)
legend("topright", legend = c("Cluster 1", "Cluster 2"),
       col = colors, pch = 19)
```



b.

```
my_kmeans <- function(x, K, max_iter = 100, initial = NULL) {
  x <- as.matrix(x)
  if (!is.numeric(x) || anyNA(x)) stop("x must contain numeric values without NA")
  if (K < 1 || K > nrow(x)) stop("K must be between 1 and the number of rows")

  if (is.null(initial)) {
    assignments <- sample(rep(seq_len(K), length.out = nrow(x)))
  } else {
    assignments <- initial
    if (length(assignments) != nrow(x) ||
        any(!assignments %in% seq_len(K)) ||
```

```
        !all(seq_len(K) %in% assignments)) {
      stop("initial must assign every row to one of K nonempty clusters")
    }
  }

  means <- matrix(NA_real_, nrow = K, ncol = ncol(x))
  colnames(means) <- colnames(x)

  for (iteration in seq_len(max_iter)) {
    for (k in seq_len(K)) {
      members <- x[assignments == k, , drop = FALSE]
      if (nrow(members) > 0) {
        means[k, ] <- colMeans(members)
      }
    }

    distance2 <- matrix(NA_real_, nrow = nrow(x), ncol = K)
    for (k in seq_len(K)) {
      distance2[, k] <- rowSums(sweep(x, 2, means[k, ], "-")^2)
    }
    new_assignments <- max.col(-distance2, ties.method = "first")
    if (all(new_assignments == assignments)) break
    assignments <- new_assignments
  }

  for (k in seq_len(K)) {
    members <- x[assignments == k, , drop = FALSE]
    if (nrow(members) > 0) means[k, ] <- colMeans(members)
  }

  wcss <- sum(rowSums((x - means[assignments, , drop = FALSE])^2))
  list(assignments = assignments, means = means, wcss = wcss)
}
```

c. The two runs produce the same grouping and objective value, because the two runs produced the same grouping (allowing for cluster labels 1 and 2 to be swapped) and the same within-cluster sum of squares. K-means repeatedly assigns points to the nearest cluster mean and recalculates those means, but other starting assignments could converge to a different grouping and objective value.

```
start1 <- initial
set.seed(43212)
start2 <- sample(rep(1:2,length.out = nrow(x)))

fit1 <- my_kmeans(x, K = 2, initial = start1)
fit2 <- my_kmeans(x, K = 2, initial = start2)

fit1$assignments
```

```
##  [1] 2 1 1 2 1 2 1 1 1 1
```

```
fit1$means
```

```
##              X1         X2
## [1,] 0.06223016 -0.7962697
## [2,] 0.95396539  1.4085533
```

```
fit1$wcss
```

```
## [1] 11.90389
```

```
fit2$assignments
```

```
##  [1] 1 2 2 1 2 1 2 2 2 2
```

```
fit2$means
```

```
##              X1         X2
## [1,] 0.95396539  1.4085533
## [2,] 0.06223016 -0.7962697
```

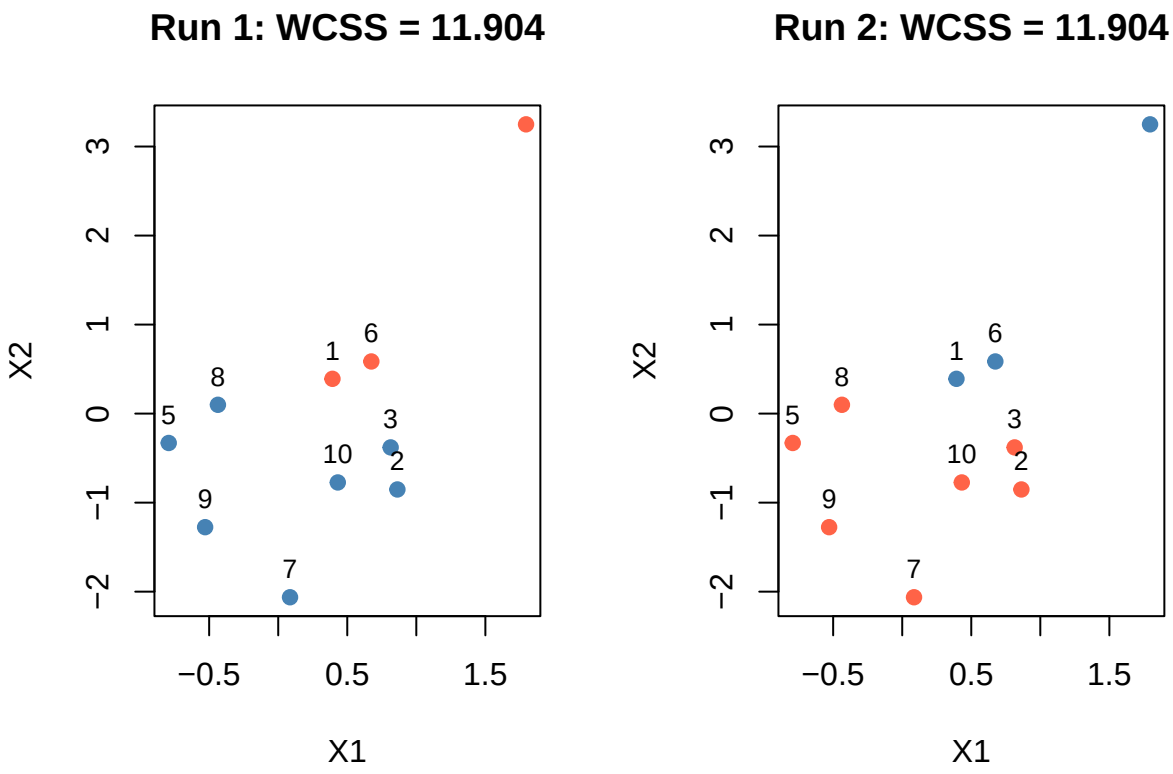
```
fit2$wcss
```

```
## [1] 11.90389
```

```
colors <- c("steelblue", "tomato")
par(mfrow = c(1, 2))

plot(x, col = colors[fit1$assignments], pch = 19,
     xlab = "X1", ylab = "X2",
     main = paste("Run 1: WCSS =", round(fit1$wcss, 3)))
text(x, labels = 1:nrow(x), pos = 3, cex = 0.8)

plot(x, col = colors[fit2$assignments], pch = 19,
     xlab = "X1", ylab = "X2",
     main = paste("Run 2: WCSS =", round(fit2$wcss, 3)))
text(x, labels = 1:nrow(x), pos = 3, cex = 0.8)
```




```
par(mfrow = c(1, 1))
```

Question 2

a.

```
image_path <- "data/illustration-42.png"  # Relative to Hw11.Rmd when knitting
img <- png::readPNG(image_path)

dim(img)
```

```
## [1] 2050 2700    4
```

```
h0 <- dim(img)[1]
w0 <- dim(img)[2]
shrink <- min(1, 200 / h0, 200 / w0)

h <- max(1, round(h0 * shrink))
w <- max(1, round(w0 * shrink))
rows <- round(seq(1, h0, length.out = h))
cols <- round(seq(1, w0, length.out = w))

small <- img[rows, cols, 1:3, drop = FALSE]
dim(small)
```

```
## [1] 152 200   3
```

```
plot(as.raster(small), main = "Original image, resized for analysis")
```



```
pixels <- cbind(
  R = as.vector(small[, , 1]),
  G = as.vector(small[, , 2]),
  B = as.vector(small[, , 3])
)
dim(pixels)
```

```
## [1] 30400     3
```

b.

```
# when k = 2
set.seed(43212)
fit2 <- my_kmeans(pixels, K = 2)
fit2$means
```

```
##              R         G         B
## [1,] 0.9795165 0.9802379 0.9833457
## [2,] 0.3394544 0.3560566 0.4082590
```

```
fit2$wcss
```

```
## [1] 722.3073
```

```
replacement_colors <- fit2$means[fit2$assignments, , drop = FALSE]
compressed2 <- array(NA_real_, dim = dim(small))
for (channel in 1:3) {
  compressed2[, , channel] <- matrix(
    replacement_colors[, channel],
    nrow = dim(small)[1],
    ncol = dim(small)[2]
  )
}

par(mfrow = c(1,2))
plot(as.raster(small), main = "Original")
plot(as.raster(compressed2), main = "K = 2")
```





```
par(mfrow = c(1, 1))
```

```
# when k = 3
set.seed(43213)
fit3 <- my_kmeans(pixels, K = 3)
fit3$means
```

```
##              R         G         B
## [1,] 0.9903701 0.9906383 0.9920004
## [2,] 0.2354519 0.2519631 0.3313273
## [3,] 0.6198894 0.6345307 0.6494434
```

```
fit3$wcss
```

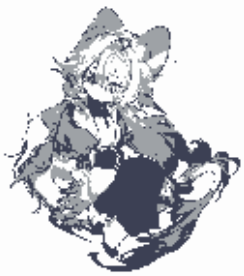
```
## [1] 330.0378
```

```
replacement_colors3 <- fit3$means[fit3$assignments, , drop = FALSE]

compressed3 <- array(NA_real_, dim = dim(small))
for (channel in 1:3) {
  compressed3[, , channel] <- matrix(
    replacement_colors3[, channel],
    nrow = dim(small)[1],
    ncol = dim(small)[2]
  )
}

par(mfrow = c(1, 2))
plot(as.raster(small), main = "Original")
plot(as.raster(compressed3), main = "K = 3")
```





```r
par(mfrow = c(1, 1))
```

```r
# when k = 5
set.seed(43215)
fit5 <- my_kmeans(pixels, K = 5)
fit5$means
```

```
##              R         G         B
## [1,] 0.8100214 0.8185685 0.8521473
## [2,] 0.9989734 0.9988300 0.9986674
## [3,] 0.3178845 0.3371809 0.4186211
## [4,] 0.1375453 0.1500853 0.2233189
## [5,] 0.5728277 0.5891137 0.5880391
```

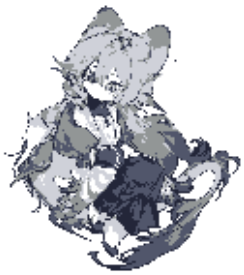
```r
fit5$wcss
```

```
## [1] 127.0261
```

```r
replacement_colors5 <- fit5$means[fit5$assignments, , drop = FALSE]
compressed5 <- array(NA_real_, dim = dim(small))
for (channel in 1:3) {
  compressed5[, , channel] <- matrix(
    replacement_colors5[, channel],
    nrow = dim(small)[1],
    ncol = dim(small)[2]
  )
}

par(mfrow = c(1,2))
plot(as.raster(small), main = "Original")
plot(as.raster(compressed5), main = "K = 5")
```





```
par(mfrow = c(1, 1))
```

```
# when k = 10
set.seed(432110)
fit10 <- my_kmeans(pixels, K = 10)
fit10$means
```

```
##               R         G         B
##  [1,] 0.3893569 0.4028592 0.4466915
##  [2,] 0.9344740 0.9317857 0.9344185
##  [3,] 0.6078179 0.6534140 0.3580985
##  [4,] 0.2366470 0.2595725 0.3814681
##  [5,] 0.6813422 0.6935320 0.7465122
##  [6,] 0.9629713 0.9284314 0.8002262
##  [7,] 0.5355235 0.5459039 0.6014083
##  [8,] 0.9999217 0.9998884 0.9998386
##  [9,] 0.8260328 0.8350323 0.8694191
## [10,] 0.1145414 0.1235125 0.1803922
```

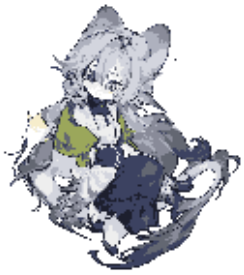
```
fit10$wcss
```

```
## [1] 54.10098
```

```r
replacement_colors <- fit10$means[fit10$assignments, , drop = FALSE]
compressed10 <- array(NA_real_, dim = dim(small))
for (channel in 1:3) {
  compressed10[, , channel] <- matrix(
    replacement_colors[, channel],
    nrow = dim(small)[1],
    ncol = dim(small)[2]
  )
}

par(mfrow = c(1,2))
plot(as.raster(small), main = "Original")
plot(as.raster(compressed10), main = "K = 10")
```





```r
par(mfrow = c(1, 1))
```

c. I will choose k = 5. Compared to k = 2 and k = 3, k = 5 has enough information needed to recover the image, but is not as complex as k = 10. When k = 3, some details may not show, and that's one reason I insist 5 is better though choosing a smaller k is generally better.

Question 3

a. In the single graph, cars Ford Pantra, Maserati Bora are joined early, with car Chrystler Imperial seperated out with other cars, while in the complete dendrogram, the cars are more averaged out,

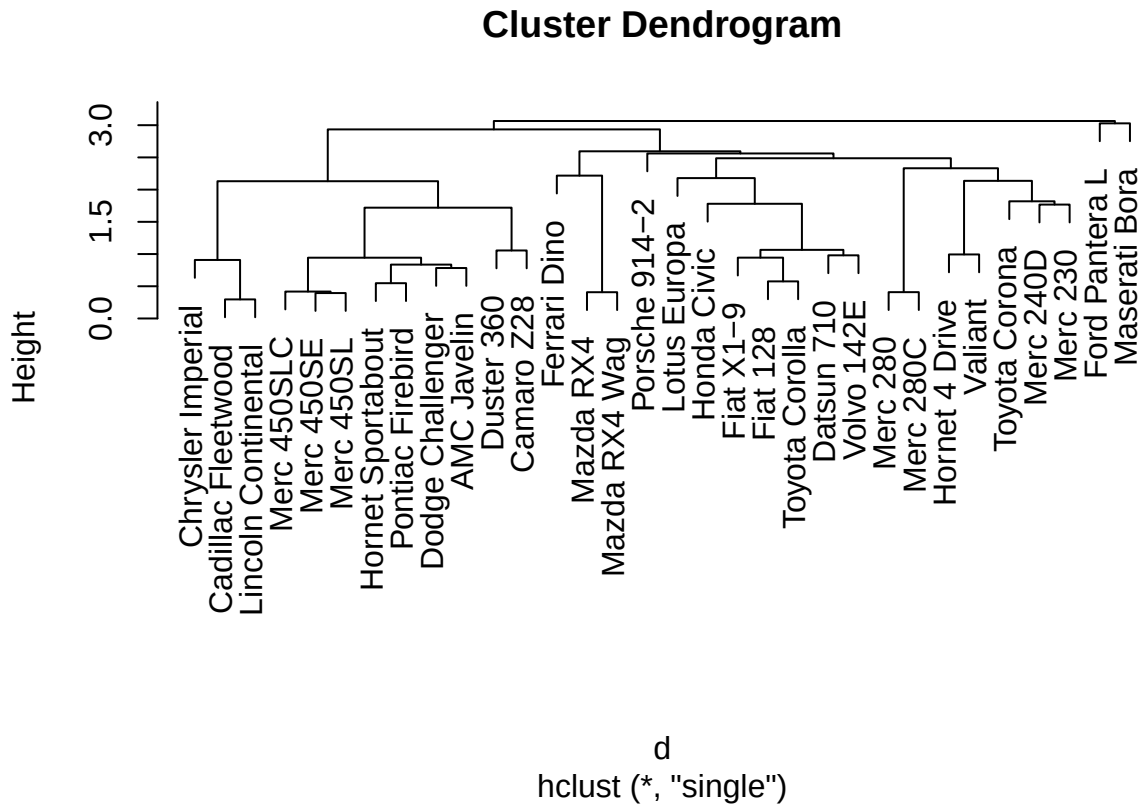
clustering into two main groups. In the average dendrogram, car Chrystler Imperial is also seperated out.

```
mtcars <- read.csv("/Users/zhijinghuang/Downloads/Stat432_2026/week-11/data/mtcars.csv")
car_names <- mtcars$car
car_variables <- mtcars[, names(mtcars) != "car"]

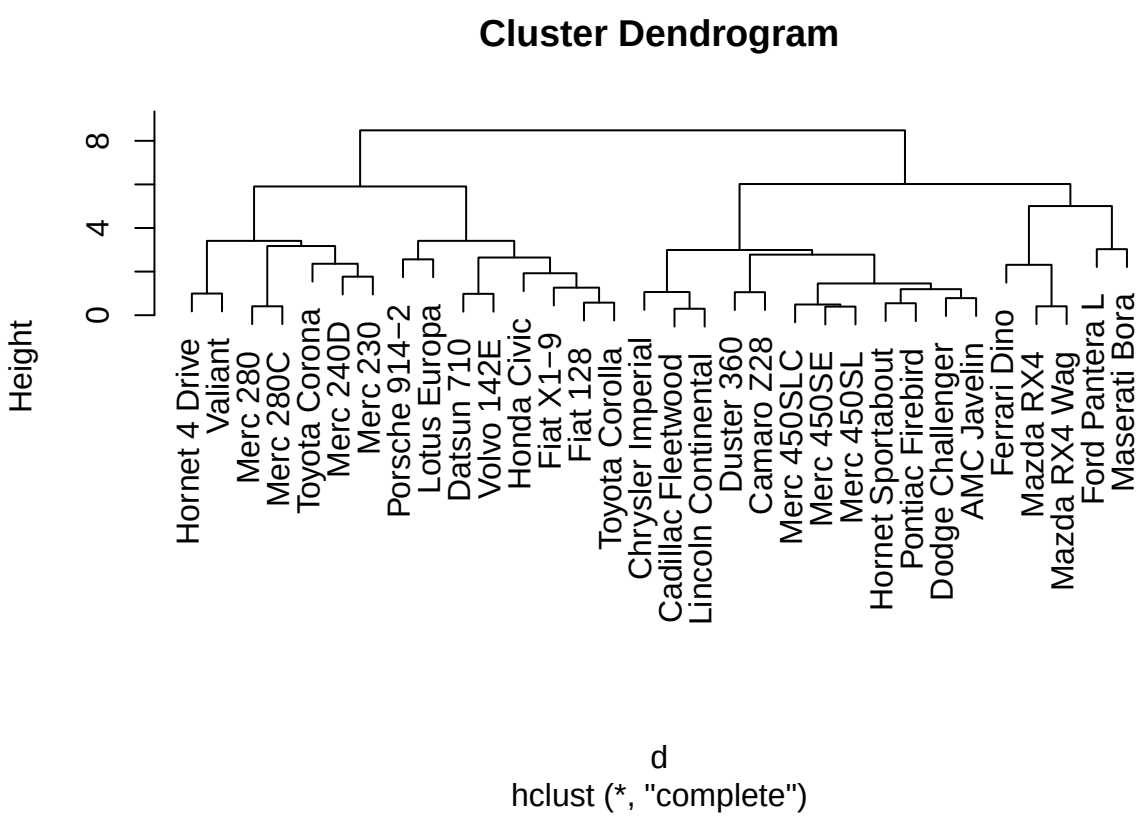
z <-scale(car_variables)
rownames(z) <- car_names
d <- dist(z,method = "euclidean")

hc_single <- hclust(d, method = "single")
hc_complete <- hclust(d, method = "complete")
hc_average <- hclust(d, method = "average")

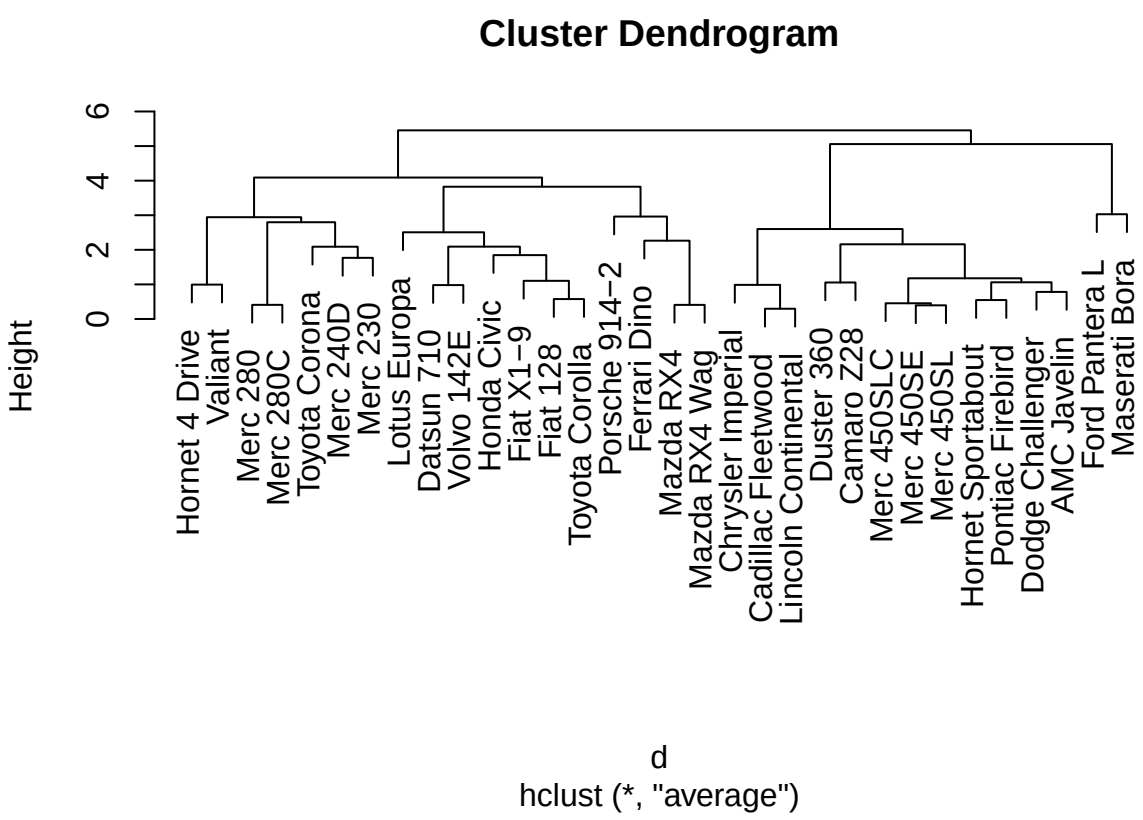
plot(hc_single)
```




```
plot(hc_complete)
```




```
plot(hc_average)
```


Cluster Dendrogram
Height
0
2
4
6
Hornet 4 Drive
Valiant
Merc 280
Merc 280C
Toyota Corona
Merc 240D
Merc 230
Lotus Europa
Datsun 710
Volvo 142E
Honda Civic
Fiat X1−9
Fiat 128
Toyota Corolla
Porsche 914−2
Ferrari Dino
Mazda RX4
Mazda RX4 Wag
Chrysler Imperial
Cadillac Fleetwood
Lincoln Continental
Duster 360
Camaro Z28
Merc 450SLC
Merc 450SE
Merc 450SL
Hornet Sportabout
Pontiac Firebird
Dodge Challenger
AMC Javelin
Ford Pantera L
Maserati Bora
d
hclust (*, "average")

b. I chose the complete linkage, and it gives three clusters, with 5 cars in the first cluster, 15 cars in the second cluster, and 12 cars in the third cluster. The five-car group includes the Mazda RX4 models and three higher-performance cars; all five have manual transmissions. I standardized the variables before clustering so variables measured on larger numerical scales would not dominate Euclidean distance.

```
groups <- cutree(hc_complete, k = 3)

split(car_names, groups)
```

```
## $`1`
## [1] "Mazda RX4"      "Mazda RX4 Wag"  "Ford Pantera L" "Ferrari Dino"
## [5] "Maserati Bora"
##
## $`2`
##  [1] "Datsun 710"     "Hornet 4 Drive" "Valiant"        "Merc 240D"
##  [5] "Merc 230"       "Merc 280"       "Merc 280C"      "Fiat 128"
##  [9] "Honda Civic"    "Toyota Corolla" "Toyota Corona"  "Fiat X1-9"
## [13] "Porsche 914-2"  "Lotus Europa"   "Volvo 142E"
##
## $`3`
##  [1] "Hornet Sportabout"   "Duster 360"          "Merc 450SE"
##  [4] "Merc 450SL"          "Merc 450SLC"         "Cadillac Fleetwood"
##  [7] "Lincoln Continental" "Chrysler Imperial"   "Dodge Challenger"
## [10] "AMC Javelin"         "Camaro Z28"          "Pontiac Firebird"
```

```
aggregate(car_variables, by = list(cluster = groups), FUN = mean)
```

```
##   cluster      mpg      cyl     disp        hp     drat       wt    qsec
## 1       1 18.50000 6.800000 223.4000 198.80000 3.836000 3.001000 15.6160
## 2       2 24.65333 4.533333 131.6467  91.33333 3.897333 2.579867 19.1580
## 3       3 15.05000 8.000000 357.6167 194.16667 3.120833 4.104083 17.1425
##          vs        am     gear     carb
## 1 0.0000000 1.0000000 4.600000 5.200000
## 2 0.9333333 0.5333333 3.933333 1.800000
## 3 0.0000000 0.0000000 3.000000 3.083333
```